$$= \left[1 + 1 + \frac{(1)(2)}{6}\right]$$

$$\therefore \int_1^2 x^2\, dx = \frac{7}{3}$$

Exercise 2.11

Evaluate the following integrals as the limit of the sum:

1. $\int_0^1 (x+4)\, dx$
2. $\int_1^3 x\, dx$
3. $\int_1^3 (2x+3)\, dx$
4. $\int_0^1 x^2\, dx$

Exercise 2.12

Choose the correct answer:

1. $\int \frac{1}{x^3} dx$ is

   (a) $\frac{-3}{x^2} + c$ (b) $\frac{-1}{2x^2} + c$

   (c) $\frac{-1}{3x^2} + c$ (d) $\frac{-2}{x^2} + c$

2. $\int 2^x dx$ is

   (a) $2^x \log 2 + c$ (b) $2^x + c$

   (c) $\frac{2^x}{\log 2} + c$ (d) $\frac{\log 2}{2^x} + c$

3. $\int \frac{\sin 2x}{2\sin x} dx$ is

   (a) $\sin x + c$ (b) $\frac{1}{2}\sin x + c$

   (c) $\cos x + c$ (d) $\frac{1}{2}\cos x + c$

4. $\int \frac{\sin 5x - \sin x}{\cos 3x} dx$ is

   (a) $-\cos 2x + c$ (b) $-\cos 2x + c$

   (c) $-\frac{1}{4}\cos 2x + c$ (d) $-4\cos 2x + c$

5. $\int \frac{\log x}{x} dx$, $x > 0$ is

   (a) $\frac{1}{2}(\log x)^2 + c$ (b) $-\frac{1}{2}(\log x)^2$

   (c) $\frac{2}{x^2} + c$ (d) $\frac{2}{x^2} + c$

6. $\int \frac{e^x}{\sqrt{1+e^x}} dx$ is

   (a) $\frac{e^x}{\sqrt{1+e^x}} + c$ (b) $2\sqrt{1+e^x} + c$

   (c) $\sqrt{1+e^x} + c$ (d) $e^x\sqrt{1+e^x} + c$

7. $\int \sqrt{e^x}\, dx$ is

   (a) $\sqrt{e^x} + c$ (b) $2\sqrt{e^x} + c$

   (c) $\frac{1}{2}\sqrt{e^x} + c$ (d) $\frac{1}{2\sqrt{e^x}} + c$

8. $\int e^{2x}[2x^2 + 2x] dx$

   (a) $e^{2x}x^2 + c$ (b) $xe^{2x} + c$

   (c) $2x^2e^2 + c$ (d) $\frac{x^2e^x}{2} + c$

9. $\int \frac{e^x}{e^x + 1} dx$ is

   (a) $\log\left|\frac{e^x}{e^x+1}\right| + c$ (b) $\log\left|\frac{e^x+1}{e^x}\right| + c$

   (c) $\log|e^x| + c$ (d) $\log|e^x + 1| + c$

10. $\int \left[\frac{9}{x-3} - \frac{1}{x+1}\right] dx$ is

    (a) $\log|x-3| - \log|x+1| + c$

    (b) $\log|x-3| + \log|x+1| + c$

    (c) $9\log|x-3| - \log|x+1| + c$

    (d) $9\log|x-3| + \log|x+1| + c$

11. $\int \frac{2x^3}{4+x^4} dx$ is

    (a) $\log|4+x^4| + c$ (b) $\frac{1}{2}\log|4+x^4| + c$

    (c) $\frac{1}{4}\log|4+x^4| + c$ (d) $\log\left|\frac{2x^3}{4+x^4}\right| + c$

12. $\int \frac{dx}{\sqrt{x^2-36}}$ is

    (a) $\sqrt{x^2-36} + c$

    (b) $\log\left|x + \sqrt{x^2-36}\right| + c$

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